

that can damage a satellite or spacecraft. According to experts, the problem will get worse. By 2025 as many as 1100 satellites could be launching each year. The number of satellites orbiting Earth is projected to quintuple over the next decade.

The estimated 8800 tons of objects that humans have left in space are becoming a danger. Near misses are common these days. In September 2019, there was one near miss between Elon Musk's SpaceX satellite and one from the European Space Agency.

But so far, there has been just one major collision. In 2009, American satellite Iridium 33 and Cosmos 2251, a Russian satellite, crashed, destroying both over northern Siberia. In January 2020, a satellite run by AT&T's DirecTV was found to be in danger of exploding and needed to be moved, or else it could harm other satellites.

In April 2020, the FCC of USA voted to require more disclosures from satellite operators seeking licenses but declined to introduce any new laws governing the removal of orbital debris. Space debris is emerging one of the main threats for an affordable and safe space exploration and exploitation.

Background

One may ask, "What is orbital debris?" Though one doesn't see space junk in the sky, beyond the clouds and further than the eye can see, it enters Low Earth Orbit (LEO), which has become an orbital space junk yard. There are millions of pieces of space junk flying in LEO. Most orbital debris comprises human-generated objects, such as pieces of spacecraft, tiny flecks of paint from a spacecraft, parts of rockets, satellites that are no longer working, or explosions of objects in orbit flying around in space at high speeds.

Most space junk is moving very fast and can reach speeds of



Fig. 2: Lockheed Martin has announced the construction of a Space Fence on the Marshall Islands in the central Pacific Ocean to track and identify space objects

28,000km per hour, almost seven times faster than a bullet. Due to the rate of speed and volume of debris in LEO, current and future space based services, explorations, and operations pose a safety risk to people and property in space and on Earth.

There are no international space laws to clean up debris in LEO. It's very expensive to remove space debris. The NASA Orbital Debris Program officially began in 1979 in the Space Sciences Branch at the Johnson Space Center (JSC) in Houston, Texas. The program looks for ways to create less orbital debris, and designs equipment to track and remove the debris already in space.

In coming years, the number of satellites will increase by an order of magnitude, with multiple mega-constellations made up of hundreds or even thousands of satellites planned for LEO to deliver wide-coverage, low-latency telecommunications, and monitoring services. Space junk is no one country's responsibility, but the responsibility of every space-faring country. The space around our planet is filled with rubbish. It's time to take out the trash!

Innovative strategies

Lockheed Martin announced the construction of a Space Fence on

the Marshall Islands in the central Pacific Ocean to track and identify space objects. Astroscale, a private company in Japan, whose mission is to clean up such space debris, is leading the charge to clean up our space pathways. The Japanese company is currently working with Japan's Aerospace Exploration Agency (JAXA) to carry out the agency's Commercial Removal of Debris Demonstration (CRD2) project.

The JAXA mission plans to complete its first phase by the end of 2022. The goal of the mission is to launch a satellite that will observe and acquire data on the rocket upper stage that the second phase will seek to deorbit. The idea is to find out how the debris moves in space and set up a safe and successful removal.

Northrop Grumman launched its first Mission Extension Vehicle spacecraft (MEV-1) in 2019 to prove it could intercept falling satellites, remove them from traffic, repair them, and put them back in orbit. Swiss start-up ClearSpace, meanwhile, has a more specific goal to remove a 100kg Vega Secondary Payload Adapter (Vespa) upper-stage rocket orbiting around 645km (400 miles) above Earth.

It plans to do that in 2025.